Soft vs. Hard Constraints in One-Dimensional Heat PINNs

A paper-aligned analytical test and a five-seed controlled extension

Abstract

We compare soft loss-based imposition of initial and boundary conditions with a hard output transformation in physics-informed neural networks (PINNs). We implement an analytical no-source heat-equation example from Bowman et al. (2023), then test a manufactured-source benchmark under five paired random seeds. Hard constraints reduce relative errors in both tested settings. This is a partial paper-aligned reproduction, not a replication of the authors' nonpublic PAC1D simulations.

1. Governing equation and constraint formulations

We solve $u_t - \alpha u_{xx} = Q(x,t)$ on x,t in $[0,1]$, with $\alpha = 0.1$, $u(x,0) = \sin(\pi x)$, and $u(0,t) = u(1,t) = 0$. Physics loss is the mean squared PDE residual, computed using PyTorch automatic differentiation.

Method	Network output	Training objective
Soft PINN	$u = N(x,t)$	$L_{PDE} + \lambda_{IC} L_{IC} + \lambda_{BC} L_{BC}$
Hard PINN	$u = \sin(\pi x) + t\, x\, (1 - x)\, N(x,t)$	L_{PDE} only

The Hard transformation satisfies the specified initial and Dirichlet boundary conditions for all network parameters (up to floating-point accuracy). It does not automatically satisfy the PDE. Within a paired experiment, raw MLP initialization and collocation samples are matched.

2. Experimental design

Parameter	Paper-aligned analytic case	Manufactured-source extension
Exact reference	$\exp(-\alpha\pi^2 t)\sin(\pi x)$	$\exp(-t)\sin(\pi x)+0.2t\sin(3\pi x)$
Forcing / source	$Q = 0$	$Q = u^*_t - \alpha\, u^*_{xx}$
MLP	10 hidden layers x 32, Tanh	3 hidden layers x 64, Tanh
Training	2,000 Adam + 120 L-BFGS steps	2,000 Adam epochs
Sampling	25k LHS pool; 125 IC; 175 BC	256 PDE / 128 IC / 128 BC per epoch
Replications	one paired seed (11)	five paired seeds (1-5)

The analytical reference is used only to determine the prescribed forcing (when present) and to evaluate test errors; interior exact temperatures are not fitted as supervised training targets.

3. Paper-aligned analytic reproduction and hard extension

Bowman et al. (2023), Eq. (12), give $u^*(x,t) = \exp(-\alpha\pi^2 t)\sin(\pi x)$ for the no-source heat equation. We reproduce the corresponding analytical solution subcase with a width-32, 10-hidden-layer Tanh PINN and a 21:1 IC/BC-to-PDE penalty ratio. The Soft and Hard cases share the same training setup and are evaluated on a 101 x 101 grid and 2,048 unseen collocation points.

Metric (seed 11)	Soft PINN	Hard PINN
Relative L2 error	0.330254%	0.008373%
Held-out PDE residual RMSE	0.019642	0.000802
Initial-condition RMSE	0.001812	0
Boundary-condition RMSE	0.001487	~6.18e-8
R2 test score	0.9999577	0.99999997

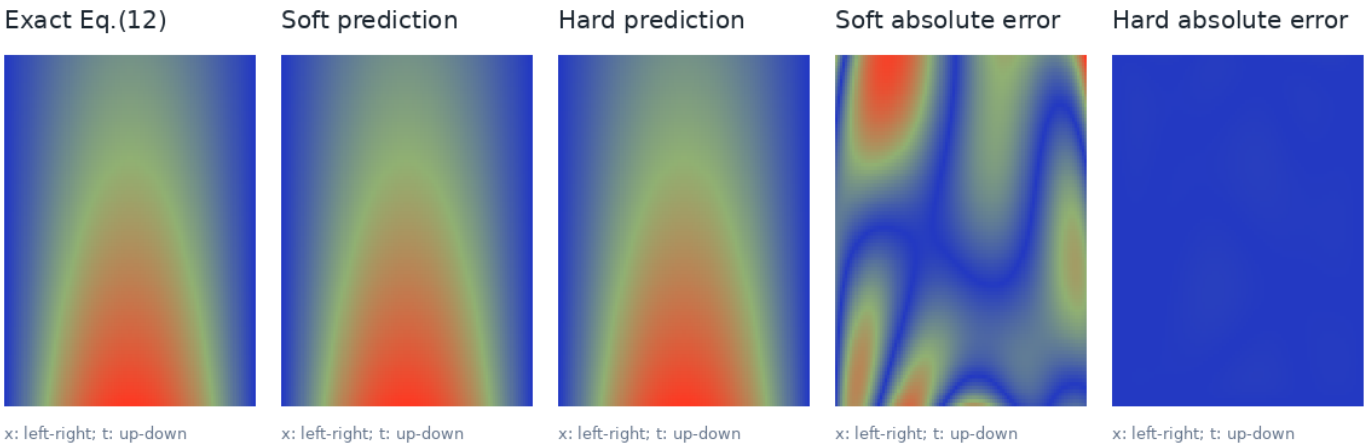


Figure 1. Exact analytic temperature, Soft and Hard predictions, and absolute errors. The two error panels share one error-color scale.

Interpretation and deviations from the paper

Hard enforcement reduces relative L2 error by 97.5% for this single matched pair and lowers the held-out PDE residual. The original study reports a 25,000-point collocation set and uses L-BFGS. Here we use a 25,000-point Latin-hypercube candidate pool, stochastic 256-point Adam updates for 2,000 steps, and then 120 nominal L-BFGS steps on 1,024 fixed collocation points. We select nondimensional $\alpha = 0.1$ rather than reproducing the original material diffusivity.

Boundary-condition clarification: Although the paper calls its no-source example "insulated," Eq. (12) has nonzero spatial gradients at x=0 and x=1. We therefore use its analytically compatible Dirichlet temperature traces; this does not replicate an insulated zero-flux problem. The PAC1D-generated source-study data were unavailable for public download. Accordingly, our results must not be compared numerically to the paper's original PAC1D results.

4. Manufactured-source extension: five matched seeds

We construct the known forcing from u*(x,t)=exp(-t) sin(πx) + 0.2t sin(3πx) and compare a 3 x 64 Tanh Soft PINN (λ_IC = λ_BC = 10) with its Hard counterpart. Both are trained for exactly 2,000 Adam epochs at learning rate 10^-3. Five paired random seeds use matching raw MLP parameters and interior training points. The figure and table summarize executed experiments, not planned configurations.

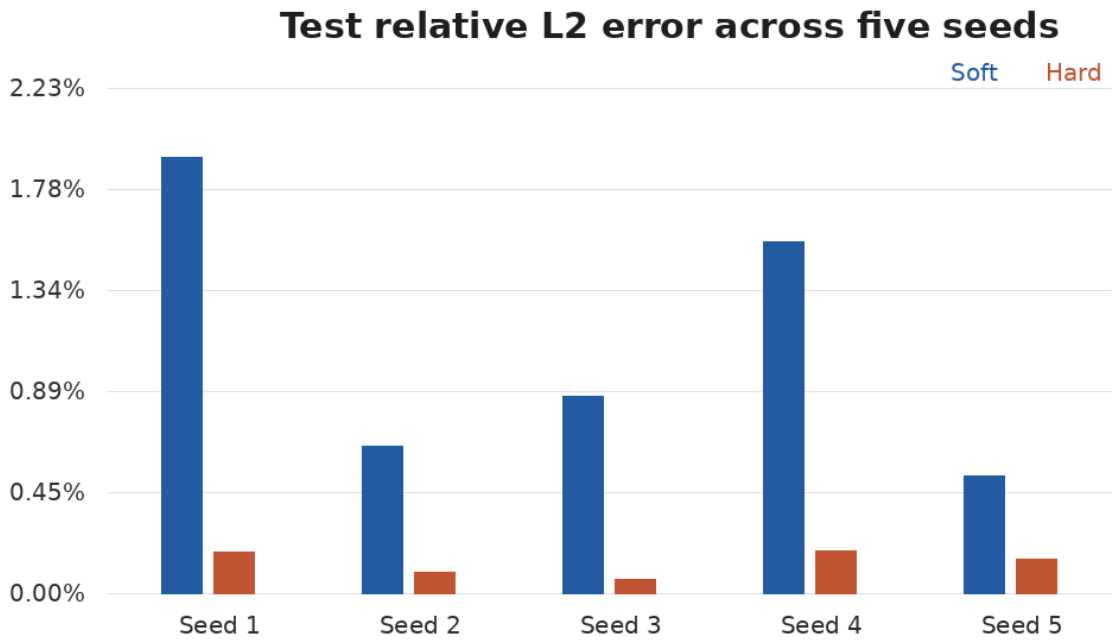


Figure 2. Relative L2 error on the same held-out evaluation grid for each paired seed.

Metric, mean +/- sample SD (n=5)	Soft	Hard
Relative L2 test error	1.1009 +/- 0.6064%	0.1331 +/- 0.0559%
Held-out PDE residual RMSE	0.03582 +/- 0.00686	0.00659 +/- 0.00223
Initial-condition RMSE	0.004312 +/- 0.002161	0
Boundary-condition RMSE	0.005574 +/- 0.002553	~6.18e-8

Conclusion, limitations and reproducibility

Hard PINN is more accurate in all five paired runs; mean relative L2 error is 87.9% lower, with less seed-to-seed variation. Its lower held-out PDE RMSE shows that this benefit is not restricted to boundary satisfaction. These results only apply to the selected 1D Dirichlet benchmarks and fixed Soft weights. We have not compared tuned/adaptive Soft penalties, Neumann/Robin constraints, or full laser-tissue PAC1D data. Full scripts, checkpoints, 2,000-epoch histories, reference grids and metrics accompany this report.

Reference: Bowman, B., Oian, C., Kurz, J., Khan, T., Gil, E., & Gamez, N. (2023). Physics-Informed Neural Networks for the Heat Equation with Source Term under Various Boundary Conditions. Algorithms, 16(9), 428. https://doi.org/10.3390/a16090428